Bayesian Optimisation of a Bio-Based Polyester Resin for Cured Thermoset Strength

7th July 2025

Thomas Højlund-Dodd, James Elliott

Abstract

This paper will focus on particular group of thermosets, namely, polyester resins. These liquid resins generally cured to form the matrix that surrounds reinforcing materials within the structure of composite materials such as glass-reinforced plastic. This paper initially elucidates the method by which an entirely bio-based polyester resin was manufactured, from simple organic acids and alcohols. Subsequently, a description of the method and associated results of Bayesion optimisation routines designed to improve the strengths exhibited by these materials will be brought to bear. The most salient findings include the mere possibility that these prepolymers can be manufactured simply and cheaply. Additionally, it is worth noting their strength being comparable after optimisation to their contemporary incumbents; the styrene-based unsaturated polyester resins.

1 Introduction

Plastics encompass a plethora of polymeric materials which provide a wide-range of desirable material properties; and given this, have seen use in a variety of applications, spanning both commodity and engineering settings [17]. Since the 1950s, the annual mass of plastics manufactured has grown from 2 Mt yr^{-1} [16] to around 460 Mt yr^{-1} in 2019 [12]; naturally, plastic waste tracked this rise. In 2019, 353 Mt of plastic waste was generated globally, with 49% landfilled, 22% mismanaged, 29% incinerated, and 9% recycled [12]. Mismanaged plastics may degrade and pollute watercourses with microplastics [19], incinerated plastics provide a means by which fossil carbon may reach the atmosphere [13], and sluggish development of recycling globally makes this the least likely fate of waste plastic [12]; these are a subset of the many challenges born out of the use of this family of materials.

Since around 85% of virgin plastic manufactured is thermoplastic they have generally dominated plastic sustainability literature, with papers focussed upon improvement of recycling methods [20], mitigation of microplastic waste [33], and biodegradability [32]. Despite their proportional minority, this paper focusses upon the group of plastics called thermosets- plastics characterised by cross-linked network structures, high dimensional stability, and chemical corrosion resistance [28]. It is generally impractical to depolymerise cross-linked thermoset materials which has lead to the common practice of either landfilling or incinerating these materials at end-of-life [50, 8, 48]. Contemporary European legislation has prioritised incineration ahead of landfill for composite materials with thermosetting matrices, and incoming legislation (e.g. Directives '99/31/EC' and '2000/53/EC') is likely to increase the proportion incinerated [38]. As the majority of thermoset plastics are derived from fossil sources of carbon, their combustion releases the greenhouse gas CO_2 into the atmosphere, perturbing the short-term carbon cycle [25]. This research directly treated this challenge through the application of Bayesian optimisation in the development of strength in bio-based alternatives to fossil incumbents.

Many thermosets exist including (but not limited to) phenolic resins, unsaturated polyester resins, epoxy resins, and polyurethanes [18]. A prominent group are the unsaturated polyester resins, which are routinely used in the manufacture of composite components; these resins exhibit desirable properties including ease of handling in liquid form, rapid curing, and excellent dimensional stability [3, 22]. The simplest form of conventional unsaturated polyester resin would be that formed from the mixture of poly(propylene glycol fumarate) copolymer (polymerised from propylene glycol and maleic anhydride) with a styrene reactive diluent [1]. Unfortunately, all monomers associated with conventional unsaturated polyester resins are fossil-derived products: propylene glycol is industrially manufactured from propylene oxide- a downstream chemical product of propane extracted from natural gas [25]. Maleic anhydride is synthesised by catalytic oxidation of the chemical butane [14]- this derived from fractional distillation of petroleum crude [21]. And styrene is produced from the dehydrogenation of ethylbenzene- this being the product of a reaction between benzene and ethylene, both of which are fossil-fuel derived feedstocks [23, 25].

Recently, literature has explored development of bio-based synthesis of these monomers; it has been shown that propylene glycol can be synthesised from the biodiesel-manufacturing byproduct glycerol via hydrogenolysis [11, 39], or from the hydrogenation of the sugar-fermentation derived product lactic acid [6, 51]. Meanwhile, bio-based maleic anhydride can be synthesised from either oxidehydration of n-butanol (from biomass fermentation of C5 and C6 sugars) [37] or catalytic oxidation of furfural (from the dehydration of C6 sugars) [27, 47]. Even bio-based styrene has been successfully manufactured via a biosynthetic route from glucose [30].

This paper focusses upon bio-based unsaturated polyester resins based on itaconic acid, a simple bio-based difunctional carboxylic acid manufactured by fungal fermentation of sugars by fungi such as *Aspergillus terreus* [4]. Itaconic acid's dicarboxyl and monoalkene functionality allows it to directly substitute maleic acid during polycondensation to form an unsaturated polyester [5]. Furthermore, itaconic acid can be synthesised into a variety of diesters (the family of diitaconates) which exhibit properties appropriate of a reactive diluent (i.e. radical initiated homopolymerisation and a low viscosity), allowing them to replace styrene in an eventual unsaturated polyester resin [43]. Bio-based itaconic acid has the advantage of being more directly synthesised from sugars than maleic acid, whilst diitaconates are a less volatile, less harmful and less flammable alternative reactive diluent to bio-based styrene.

Literature has explored catalytic influence upon polycondensation of itaconic acid within eventual unsaturated polyesters [41], synthesis of partially bio-based resins incorporating itaconic acid based unsaturated polyester but retaining styrene as the reactive diluent [9], as well as fully bio-based resins using itaconic acid based unsaturated polyester in tandem with diitaconates as reactive diluents [36, 35]. As an aside, literature has also considered partially bio-based resins incorporating itaconic acid based unsaturated polyesters together with the cross-linking compatible methacrylate class of reactive diluents [44, 10]. Although traditionally fossil-fuel derived, recent work has seen bio-based methyl methacrylate synthesised from itaconic acid, which would unlock fully bio-based resins based on methacrylate reactive diluents in future [34, 7]. Diitaconates were nontheless focussed upon in this research (despite inferior homopolymerisation rates [43] and viscosity characteristics [44]) ahead of methacrylates on account of their relatively higher strengths as cured thermosets [44], safety as reactive diluents (lower toxicity and flammability) [24], and environmentally-benign nature as a non-volatile organic compound [44, 49].

On account of the temporal and financial expense of synthesising the chemicals concerned, the sample-efficient technique of Bayesian optimisation was used to facilitate the development of strength in these bio-based materials [15]. Literature describes how this technique has been used to successfully accelerate optimisation in a variety of domains: improving synthesis efficiency in organic chemistry [42], reducing phenotyping costs during crop breeding in plant science [46], and tuning microfluidic device setup for polymer fibre synthesis in materials science [26].

In polymer science, Bayesian optimisation has been deployed in both purely computational domains (e.g. optimisation of thermoplastic repeat units for improved glass transition temperatures in a simulated parameter space [31]) as well as in applied practice (e.g. improving compositional homogeneity during co-polymerisation of styrene and methyl methacrylate [45]). Various thermosets have been practically optimised using Bayesian optimisation: Benzoxazine-based thermosets have been optimised with respect to six thermomechanical properties [29], polycyanurates have had three thermal/mechanical/hygroscopic properties improved upon [52], and epoxy thermosets based on amine curing agents have been optimised with respect to a trio of mechanical properties using a dual-driver method [53]. In the more specific domain of practical Bayesian optimisation of bio-based thermosets, literature has been dominated by epoxy-based chemistries; Albuquerque et al. successfully improved glass transition temperatures in bio-based epoxy thermosets derived from amino acid based resin systems [2], whilst Rothenhäusler et al. explored this class of thermosets further by taking into account both cost and bio-based content [40].

It seems that there exist no papers on practical optimisation of bio-based thermoset composites nevermind those incorporating polyester as the matrix forming material...

Can such a paper be found?

2 Method

3 Results

4 Conclusion

5 Bibliography

[1] Abdeen Ahamad et al. "Synthesis of unsaturated polyester resin- effect of anhydride composition". In: *Designed Polymers and Monomers* 4.3 (2001), pp. 261–268.

[2] Rodrigo Albuquerque et al. "Designing formulations of bio-based, multicomponent epoxy resin systems via machine learning". In: *MRS Bulletin* 49 (2024), pp. 59–70.

[3] Sharifah Aziz et al. "Modified polyester resins for natural fibre composites". In: *Composites Science and Technology* 65 (2005), pp. 525–535.

[4] Richa Bafana and R Pandey. "New approaches for itaconic acid production-bottlenecks and possible remedies". In: *Critical Reviews in Biotechnology* 38.1 (2017).

[5] P Bamane et al. "Replacement of traditional unsaturated acid by bio-based itaconic acid in the preparation of isophthalic acid-based unsaturated polyester resin". In: *Progress in Organic Coatings* 147.105743 (2020).

[6] J Berlowska et al. "A low-cost method for obtaining high-value bio-based propylene glycol from sugar beet pulp". In: *RSC Advances* 5 (2015), pp. 2299–2304.

[7] Ashish Bohre et al. "Single-step production of bio-based methyl methacrylate from biomass-derived organic acids using solid-catalyst material for cascade decarboxylation-esterification reactions". In: *ACS Sustainable Chemistry and Engineering* 9.7 (2021).

[8] Huaguo Chen et al. "Recycling thermoset plastic waste for manufacturing green cement mortar". In: *Cement and Concrete Composites* 137.104922 (2023).

[9] C Costa et al. "Going greener- synthesis of fully biobased unsaturated polyesters for styrene crosslinked resins with enhanced thermomechanical properties". In: *eXPRESS Polymer Letters* 11.11 (2017), pp. 885–898.

[10] Jinyue Dai et al. "2,5-Furandicarboxylic acid- and itaconic acid-derived fully biobased unsaturated polyesters and their cross-linked networks". In: *Industrial and Engineering Chemistry Research* 56 (2017), pp. 2650–2657.

[11] Mohanprasad Dasari et al. "Low-pressure hydrogenolysis of glycerol to propylene glycol". In: *Applied Catalysis A- General* 281.1-2 (2005), pp. 225–231.

[12] OWID (Our World in Data). *Plastic Pollution*. Ed. by Hannah Ritchie et al. Retrieved July 12th, 2025. 2025. URL: `https://ourworldindata.org/plastic-pollution`.

[13] Orla Eriksson et al. "Plastic waste as a fuel - CO2-neutral or not?" In: *Energy and Environmental Science* 2 (2009), pp. 907–914.

[14] Timothy Felthouse et al. "Maleic Anhydride, Maleic Acid, and Fumaric Acid". In: *Kirk-Othmer Encyclopedia of Chemical Technology*. John Wiley and Sons, Ltd, 2001.

[15] Peter Frazier. *A Tutorial on Bayesian Optimization*. Retrieved 28th July 2025. arXiv, 2018. eprint: `1807.02811`. URL: `https://arxiv.org/abs/1807.02811`.

[16] Roland Geyer et al. "Production, use, and fate of all plastics ever made". In: *Science Advances* 3.e1700782 (2017).

[17] Marianne Gilbert. "Brydson's Plastics Materials". In: ed. by Marianne Gilbert. 8th ed. 2017. Chap. 1.

[18] Sidney Goodman. *Handbook of Thermoset Plastics*. Ed. by Sidney Goodman. 1st ed. Noyes Publications, 1986.

[19] Hiroshan Hettiarachchi et al. "Microplastic Pollution Prevention- The Need for Robust Policy Interventions to Close the Loopholes in Current Waste Management Practices". In: *International Journal of Environmental Research and Public Health* 20.14 (2023).

[20] Praveenkumara Jagadeesh et al. "Sustainable recycling technologies for thermoplastic polymers and their composites- A review of the state of the art". In: *Polymer Composites* 43.9 (2022), pp. 5831–5862.

[21] Y Kissin. "Catagenesis and composition of petroleum- Origin of n-alkanes and isoalkanes in petroleum crudes". In: *Geochimica et Cosmochimica Acta* 51 (1987), pp. 2445–2457.

[22] John Lawrence. *Polyester Resins*. Ed. by Herbert Simonds. 1st ed. Reinhold Publishing Corporation, 1960.

[23] W Lee et al. "Ethylbenzene dehydrogenation into styrene- kinetic modeling and reactor simulation". In: *Industrial and Engineering Chemistry Research* 47.23 (2008).

[24] Edward Leonard. *Vinyl and Diene Monomers Part 1 (High Polymers XXIV) (Volume 24)*. Wiley, 1970.

[25] Peter Levi and Jonathan Cullen. "Mapping global flows of chemicals-from fossil fuel feedstocks to chemical products". In: *Environmental Science and Technology* 52.4 (2018).

[26] Cheng Li et al. "Rapid Bayesian optimisation for synthesis of short polymer fibre materials". In: *Scientific Reports* 7.5683 (2017).

[27] Xiukai Li et al. “Selective aerobic oxidation of furfural to maleic anhydride with heterogenous Mo-V-O catalysts”. In: *Green Chemistry* 18 (2016), pp. 2976–2980.

[28] Jingkai Liu et al. “Recent development on bio-based thermosetting resins”. In: *Journal of Polymer Science* 59.14 (2021), pp. 1474–1490.

[29] Yin Lu et al. “Ultrahigh performance benzoxazine/poly(silane arylacetylene) thermosetting composite materials based on smart molecular engineering and machine learning”. In: *Composites Part B: Engineering* 307.112863 (2025).

[30] Rebekah McKenna and D Nielsen. “Styrene biosynthesis from glucose by engineered E. coli”. In: *Metabolic Engineering* 13.5 (2011), pp. 544–554.

[31] Takuya Minami et al. “Prediction of repeat unit of optimal polymer by Bayesian optimization”. In: *MRS Advances* 4 (2019), pp. 1125–1130.

[32] Taofeeq Moshood et al. “Sustainability of biodegradable plastics- New problem or solution to solve the global plastic pollution?” In: *Current Research in Green and Sustainable Chemistry* 5.100273 (2022).

[33] Taiseer Nafea et al. “Status of management and mitigation of microplastic pollution”. In: *Critical Reviews in Environmental Science and Technology* 54.24 (2024), pp. 1734–1756.

[34] Koichi Nagai. “New developments in the production of methyl methacrylate”. In: *Applied Catalysis A- General* 221.1-2 (2001), pp. 367–377.

[35] Vesna Panic et al. “Simple one-pot synthesis of fully biobased unsaturated polyester resin based on itaconic acid”. In: *Biomacromolecules* 18 (2017), pp. 3881–3891.

[36] Olga Pantic et al. “The effect of glycol derivatives on the properties of bio-based unsaturated polyesters”. In: *Polymers* 14.2970 (2022).

[37] Giulia Pavarelli et al. “A new process for maleic anhydride synthesis from a renewable building block-the gas-phase oxidehydration of bio-1-butanol”. In: *ChemSusChem* 8.13 (2015), pp. 2250–2259.

[38] S Pickering. “Recycling technologies for thermoset composite materials—current status”. In: *Composites Part A- Applied Science and Manufacturing* 37.8 (2006), pp. 1206–1215.

[39] Maria Ricciardi et al. “Bio-propylene glycol as value-added product from Epicerol process”. In: *Sustainable Chemistry and Pharmacy* 6 (2017), pp. 10–13.

[40] Florian Rothenhäusler and Holger Ruckdaeschel. “Strategies for the fast optimization of the glass transition temperature of sustainable epoxy resin systems via machine learning”. In: *Journal of Applied Polymer Science* 141.21 (2024).

[41] Ina Schoon et al. “Catalyst influence on undesired side reactions in the polycondensation of fully bio-based polyester itaconates”. In: *Polymers* 9.693 (2017).

[42] Benjamin Shields et al. “Bayesian reaction optimization as a tool for chemical synthesis”. In: *Nature* 590 (2021), pp. 89–96.

[43] Lea Sollka and Karen Lienkamp. “Progress in the free and controlled radical homo- and co-polymerisation of itaconic acid derivatives- toward functional polymers with controlled molar mass distribution and architecture”. In: *Macromolecular Rapid Communications* 42.4 (2020).

[44] Pavle Spasojevic et al. "Optimization of reactive diluent for bio-based unsaturated polyester resin- a rheological and thermomechanical study". In: *Polymers* 13.2667 (2021).

[45] Shogo Takasuka et al. "Bayesian optimization of radical polymerization reactions in a flow synthesis system". In: *Science and Technology of Advanced Materials: Methods* 4.1 (2024).

[46] Ryokei Tanaka and Hiroyoshi Iwata. "Bayesian optimization for genomic selection- a method for discovering the best genotype among a large number of candidates". In: *Theoretical and Applied Genetics* 131 (2017), pp. 95–105.

[47] Siew Teong et al. "Hydroxymethylfurfural production from bioresources-past, present and future". In: *Green Chemistry* 16 (2014), pp. 2015–2026.

[48] Eva Topham. "Recycling offshore wind farms at decommissioning stage". In: *Energy Policy* 129 (2019), pp. 698–709.

[49] J Trejbal et al. "Determination of density, viscosity, and saturated vapour pressure of various itaconic acid esters". In: *International Journal of Thermophysics* 43.51 (2022).

[50] Shubham Utekar et al. "Comprehensive study of recycling of thermosetting polymer composites– Driving force, challenges and methods". In: *Composites Part B - Engineering* 207.108596 (2021).

[51] Jiefeng Xiao et al. "A novel approach for 1,2-propylene glycol production from biomass-derived lactic acid". In: *Process Biochemistry* 50.5 (2015), pp. 793–798.

[52] Xinyao Xu et al. "Efficient exploration of compositional space for high-performance copolymers via Bayesian optimization". In: *Chemical Science* 14 (2015), pp. 10203–10211.

[53] Wenlin Zhao et al. "Design of multicomponent thermosetting polymers with enhanced tensile properties through active learning". In: *Composite Science and Technology* 256.110779 (2024).